Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

- Power = Voltage ÃĹ Current

- We can find the current (I) as P / V , and then use the current and voltage to find resistance (R) using Ohm's law I = V / R

- So we know Voltage (V) = 12 V , Power (P) = 40 W

- Current I = P / V = 40 W / 12 V = 3.33 A

- Resistance R = V / I = 12 V / 3.33 A = 3.6 ohms

</think>

<answer>

E. 3.33 amps, 3.6 ohms

</answer>